llmXive Automated Review of ResearchStudio-Reel: Automate the Last Mile of Research from Paper to Poster, Video, and Blog

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several central claims regarding the performance of ResearchStudio-Reel against state-of-the-art baselines. However, the evidence provided in the text and tables fails to support these claims due to the citation of non-existent models.

Specifically, Section 5 ("Experiments") and Table 1 list "GPT-5.5", "Gemini-3.1 Pro", and "Claude-4.8 Opus" as single-shot baselines. As of the current date, these model versions do not exist in the public record (the latest public versions are GPT-4o, Gemini 1.5, and Claude 3.5). The paper's results table relies entirely on comparisons against these fabricated baselines to establish the system's superiority. Without access to the actual models used or a correction to the model names, the reported "state-of-the-art" performance cannot be verified or reproduced. This constitutes a fatal flaw in the claim-accuracy chain, as the primary evidence for the paper's central contribution is based on comparisons with entities that do not exist.

Additionally, the text claims the system wins "on 84% to 93% of papers" (Abstract, Section 5). While Table 1 provides aggregate mean scores, it does not report the per-paper win rates required to substantiate this specific percentage range. The reader is left unable to verify this specific quantitative claim from the provided data.

The bibliography also contains several citations with future dates (e.g., "efficientpostergen" 2026, "apex" 2026) which, while potentially consistent with a 2026 preprint, further complicate the verification of the "prior work" landscape. However, the non-existence of the baseline models is the primary and most severe accuracy failure.

Action items.

- [**fatal**] The paper makes several central claims regarding the performance of ResearchStudio-Reel against state-of-the-art baselines. However, the evidence provided in the text and tables fails to support these claims due to the citation of non-existent models. Specifically, Section 5 ("Experiments") and Table 1 list "GPT-5.5", "Gemini-3.1 Pro", and "Claude-4.8 Opus" as single-shot baselines. As of the current date, these model versions do not exist in the public record (the latest public versions are GPT

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear, high-quality visual teaser that effectively demonstrates the Paper2Poster output. The layout is readable, the content is legible, and it successfully conveys the system's capability to transform a paper into a structured poster.

Figure 2

The figure effectively illustrates the pipeline flow and component relationships. However, the color coding in the diagram contradicts the provided legend, as the specific tool boxes (Poster, Video, Blog) are white while the legend indicates skills should be blue.

Figure 3

The figure provides a clear high-level overview of the pipeline but suffers from a disconnect between the color-coded legend and the actual diagram elements, and lacks visual indicators for the loop's iteration bounds.

Figure 4

The figure effectively visualizes the staged-fill loop process across three stages, but the lack of a visible legend for the color-coded debug overlay and the illegibility of the fill percentage annotations hinder the reader's ability to interpret the specific verdicts and metrics.

Figure 5

The figure provides a clear visual overview of the Paper2Video pipeline, but the 'Pace narration' inputs conflict slightly with the caption's description of planning, and a small attribution text in the workflow box is unexplained.

Figure 6

Figure 6 effectively visualizes the Paper2Video deliverables and user controls. The layout clearly pairs the editable PowerPoint deck with the rendered video player, while the bottom section explicitly defines the four key configuration parameters (Duration target, Highlight styles, Caption mode, Video specs) with sufficient detail.

Figure 7

Figure 7 is a clear and well-structured pipeline diagram that effectively visualizes the workflow described in the caption. All stages, from 'Assets' to 'ARTIFACTS', are legible, and the internal logic of the 'Multi-lingual blog drafts' and 'Editorial QA loop' boxes is easy to follow without clutter.

Figure 8

The figure attempts to showcase layout checks for two documents but fails to provide clear evidence for the claims made. The annotations are not self-explanatory, and the visual evidence (e.g., cropped figure, cut-off text) contradicts the stated purpose of the checks. The Chinese document is shown without any corresponding analysis.

Figure 9

Figure 9 effectively demonstrates the Paper2Reel interaction showcase as described in the caption. The two panels clearly distinguish between the hover state (highlighting poster sections) and the double-click state (opening the synchronized modal with video and blog content), with all UI elements like language switches and slide thumbnails visible and legible.

Figure 10

The figure image appears to be a placeholder or incorrect file (a scientific poster) that does not match the caption's description of an ablation study on poster generation. Additionally, specific panels mentioned in the caption (Codex, max reasoning) are not labeled in the image.

Figure 11

The figure renders only a single poster despite the caption describing a multi-panel comparison with specific baselines; the claimed comparison is not visible.

Action items.

- [**writing**] Figure 2: The legend defines 'Skills' as light blue boxes, but the 'Paper2Assets' box is light blue while the subsequent 'Paper2Poster', 'Paper2Video', and 'Paper2Blog' boxes are white, contradicting the legend's implication that they are all skills.
- [**writing**] Figure 2: The 'Reusable Assets' box contains a 'JPEG' icon, but the caption does not explicitly list 'figures/logos' as part of the bundle, creating a minor disconnect between the visual detail and the textual description.
- [**writing**] Figure 3: The 'Staged-fill loop' legend uses color-coded labels (OVERFLOW, SPILLAGE, etc.) that are not visually represented in the diagram's flowchart elements, creating a disconnect between the legend and the process visualization.
- [**science**] Figure 3: The 'Staged-fill loop' diagram depicts a feedback cycle but lacks explicit iteration counters or stopping criteria visualization, making it difficult to verify the 'bounded ~12 rounds' claim mentioned in the caption.
- [**writing**] Figure 4: The caption defines the color coding for the debug overlay (red/amber for EMPTY/SPARSE, green for FULL, orange/magenta for SPILLAGE/OVERFLOW), but the rendered image lacks a visible legend or key to map these colors to their specific verdicts.
- [**writing**] Figure 4: The fill percentage annotations on the debug boxes are extremely small and illegible in the rendered image, making it impossible to verify the '90–98%' claim

mentioned in the caption.

- [**writing**] Figure 5: The ‘Pace narration’ box lists ‘target duration’ and ‘section ids’ as inputs, but the caption states the skill ‘plans narration and duration,’ creating ambiguity on whether these are user inputs or internal outputs.
- [**writing**] Figure 5: The ‘Run pipeline ppt-master’ box contains a small attribution ‘by Hugo He’ that is not mentioned in the caption and appears to be a watermark or credit rather than a functional diagram element.
- [**science**] Figure 8: The ‘Typography gate’ annotation points to a specific paragraph in the English document, but the text is not ‘balanced’ (it is a standard left-aligned block); the visual evidence does not support the claim of ‘balanced article lines’ or the specific location of the check.
- [**science**] Figure 8: The ‘Figure-fit gate’ annotation points to a diagram in the English document, but the diagram is clearly cropped and cut off at the bottom edge of the page, contradicting the claim that the figure is ‘sized to the flow’.
- [**science**] Figure 8: The ‘Pagination gate’ annotation points to the bottom of the English page, but the text ends abruptly in the middle of a sentence (‘...making global relationships easier to model while keeping the computation highly parallel and scalable.’), indicating a pagination error rather than a successful check.
- [**writing**] Figure 8: The annotations on the left (e.g., ‘Typography gate’, ‘Figure-fit gate’) are not defined in the figure caption or the image itself; the viewer must infer their meaning from the text below them, which is not a standard or clear way to present a ‘showcase’ of checks.
- [**writing**] Figure 8: The Chinese document (right) has no corresponding annotations or checks shown, making it impossible to verify if the same layout checks were applied to it, despite the caption stating ‘the two required Word deliverables’ are shown together with the checks.
- [**science**] Figure 10: The caption claims this is a ‘Qualitative ablation study’ varying ‘harness and base model’, but the image displays a scientific poster titled ‘Butterfly Effects of SGD Noise’ (a different paper entirely). The visual content does not match the caption’s description of the study or the artifacts being compared.
- [**science**] Figure 10: The caption references a ‘Codex panel’ and a ‘max reasoning’ panel, but the image contains no such labels or text annotations to identify these specific variants.
- [**science**] Figure 11: The caption claims to show ‘Paper2Poster Tool paper2poster’ and ‘PosterGen postergen’ baselines in the center, but the rendered image displays a single large poster with no visible baselines or comparison panels.
- [**writing**] Figure 11: The caption lists specific baselines (Paper2Poster Tool, PosterGen, P2P, and three LLMs) that are not visually present or labeled in the rendered image, making the comparison claim unverifiable.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a technical audience, but it relies on several acronyms and specific internal terminology that are not defined at their first occurrence, creating minor barriers for a competent reader from an adjacent field (e.g., a researcher in NLP or Human-Computer Interaction who is not deeply embedded in the specific "agent skills" subfield).

Specifically, the acronym "VLM" (Vision-Language Model) appears in the Abstract and Introduction without being spelled out. While standard in current ML literature, the review criteria require explicit definition for adjacent-field readers. Similarly, the symbol `fullRatio` in Section 3.2 is introduced with an equation but lacks a precise definition of its constituent variables (`h_content`, `h_card`) regarding their units or domain (e.g., CSS pixels vs. logical units).

Additionally, the paper uses specific proper nouns like "OMML" (Office Math Markup Language) and "ppt-master" as if they were common knowledge. "OMML" is a niche Microsoft standard, and "ppt-master" is a specific external tool referenced only by citation; a brief gloss explaining what these are would prevent the reader from having to look up external references to understand the pipeline's mechanics. Finally, the term "Scan-to-Read block" is used as a specific component name without a brief descriptive gloss.

These are all low-friction fixes (adding a parenthetical expansion or a short clause) that would significantly improve the self-containment of the paper without altering its technical substance.

Action items.

- [**writing**] Section 1 (Introduction) and Abstract use 'VLM' (Vision-Language Model) without expansion. While common in ML, an adjacent-field PhD (e.g., NLP or Systems) might not assume this acronym is universal. Define at first use: 'vision-language model (VLM)'.
- [**writing**] Section 3.2 (Paper2Poster) introduces the symbol `fullRatio` in the equation `fullRatio = h_content / h_card` without explicitly defining the sets or units for `h_content` and `h_card` (e.g., 'where h_content is the rendered content height in CSS pixels'). Add a brief clause defining these terms.
- [**writing**] Section 3.2 (Paper2Poster) uses the term 'OMML' (Office Math Markup Language) when describing the conversion of MathJax equations to PowerPoint. This is a specific Microsoft standard not known outside the Office ecosystem. Expand to 'Office Math Markup Language (OMML)' at first use.
- [**writing**] Section 3.2 (Paper2Poster) and Section 3.3 (Paper2Video) refer to 'ppt-master' as a workflow dependency. While a citation is provided, the term is used as a proper noun without a brief gloss of what it is (e.g., 'an AI-driven slide generation skill'). Add a one-clause definition.

- **[writing]** Section 3.2 (Paper2Poster) uses the phrase ‘Scan-to-Read block’ as a specific UI component name. While descriptive, it is introduced as a defined term without prior context. Ensure it is clear this is a specific named block in the poster layout, perhaps by adding ‘the Scan-to-Read block (a QR code section)’ on first mention.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper’s argument structure is logically sound and internally consistent. The central thesis—that a composition of five skills sharing a single upstream extractor (Paper2Assets) solves the problems of isolated extraction (G1), one-way renders (G2), and soft quality gates (G3)—is supported by a coherent chain of reasoning throughout the text.

The definitions of the five skills (Paper2Assets, Paper2Poster, Paper2Video, Paper2Blog, Paper2Reel) are established in Section 3 and used consistently in the Experiments (Section 5) and Applications (Section 6). The causal claims regarding the “measured-fill loop” improving aesthetics are supported by the ablation study in Section 5, which explicitly isolates the loop’s contribution by holding the model fixed while changing the harness. The results in Table 1 align with the textual claims: the text states the method leads on aesthetics and information sub-criteria, and the table data confirms this with bolded/underlined values for the ResearchStudio-Reel rows.

There are no contradictions between sections. The Limitations section (Appendix A) honestly addresses the gap to human posters (generative vs. compositional) and the proxy-bound nature of the evaluation, which is consistent with the Future Work section and does not undermine the main claims. The distinction between the “Claude Code” and “Codex” settings is maintained consistently in the text and tables. The numerical values in the pipeline breakdown table (Table 2) sum correctly to the reported totals, and the cost analysis logic (cached vs. fresh tokens) is applied consistently.

The argument that the system is the “only pipeline to ship all three editable artifacts” is supported by the capability audits in Tables 3 and 4, which clearly mark prior systems with ‘x’ for missing capabilities while marking the proposed system with checkmarks. The logical flow from problem definition to architectural solution to empirical validation is tight, with no non-sequiturs or unsupported leaps in the reasoning.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a compelling system for automating research dissemination, but the rhetoric in the Abstract and Conclusion occasionally outpaces the specific scope of the evidence provided.

First, the Abstract and Conclusion claim the system "surpasses the authors' own" posters on aesthetics with a specific margin. However, the "Results" section text cites numbers (3.52 vs 2.94) that contradict the primary results in Table 1 (3.33 vs 3.02). This discrepancy suggests the summary claim may be based on a different aggregation or subset not clearly defined. The Abstract should align its numerical claims with the primary results table to ensure the "surpassing" claim accurately reflects the reported evidence.

Second, the claim that ResearchStudio-Reel is "the only pipeline to ship all three editable artifacts" is framed as a universal fact. While the capability audits support this against the *specific* baselines chosen (single-shot LLMs and a selection of prior tools), the Related Work section acknowledges a wider literature. The claim should be qualified to "among the systems evaluated in this study" or "to our knowledge, the first to integrate these three specific editable formats in a single pipeline" to accurately reflect the scope of the comparative evidence.

Finally, the Conclusion asserts that "any dissemination target with a deterministic render... fits the same composition." This is a broad generalization about the architecture's applicability to *any* future artifact type. The paper only validates this composition on three specific artifact types. While the authors' intuition is sound, presenting this as a settled fact rather than a hypothesis overextends the current experimental validation. Rephrasing this as a hypothesis ("We hypothesize that this pattern generalizes to...") would better match the evidence.

These are primarily issues of framing and precision. Narrowing the scope of the claims to match the specific baselines and results presented will strengthen the paper's credibility.

Action items.

- [**writing**] Abstract/Conclusion claim 'surpasses authors' own' with specific margin (3.52 vs 2.94), but Table 1 shows 3.33 vs 3.02. The text in Sec 5 contradicts the table. Align the abstract's numerical claim with the table's reported means to avoid misrepresenting the evidence.
- [**writing**] Abstract claims the system is 'the only pipeline to ship all three editable artifacts.' This implies a universal fact, but capability audits (Tables 2-3) only compare against a curated subset of baselines. Scope the claim to 'among systems evaluated here' or 'to our knowledge' to match the evidence.
- [**writing**] Conclusion states 'any dissemination target... fits the same composition' as a fact. The paper only validates this on three artifact types. Rephrase as a hypothesis (e.g., 'We hypothesize this pattern extends to...') to avoid overgeneralizing beyond the demonstrated scope.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a system for automating the creation of research dissemination artifacts (posters, videos, blogs) from academic papers. This is a low-risk application domain. The system processes user-provided PDFs or LaTeX sources and generates editable outputs; it does not involve scraping private data, training on non-consensual human datasets, or generating content designed to deceive, surveil, or cause physical harm.

The authors have included a dedicated "Ethics" appendix (Section app:ethics) that appropriately addresses the primary risks associated with this work:

1. **Provenance and Disclosure:** The paper explicitly recommends preserving metadata to disclose AI involvement, acknowledging that generated artifacts could otherwise be mistaken for human-made work.
2. **Licensing:** The authors correctly note that figure copyright follows the source paper's license and that institution logos are fetched from public sources (Wikimedia/Wikidata) under their respective licenses. They clarify that the system does not redistribute third-party content but rather processes user-provided inputs.
3. **Hallucination/Factual Consistency:** The "Limitations" section (app:limit) and the system design (shared evidence map, hard gates) address the risk of factual drift or hallucination, which is the primary safety concern for automated summarization systems.

There are no indications of dual-use capabilities (e.g., generating disinformation at scale, impersonating specific individuals without consent, or creating deepfakes of real people). The "video" component uses text-to-speech and slide animations, not generative video of human faces. The "blog" component produces text in Word format, not deceptive social media content.

No human-subjects data, PII, or sensitive datasets are used or released. The evaluation relies on a public benchmark (Paper2Poster) and the authors' own generated outputs. The paper does not require an IRB statement as no human subjects were recruited or observed.

The risk profile is minimal, and the authors have provided adequate disclosure regarding the limitations and ethical considerations of their system. No action items are required.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling system for automating research dissemination, but the central quantitative claims regarding performance superiority over human authors and baselines rest on experimental designs that lack necessary controls for variance and confounding factors.

First, the headline result in Table 1—that ResearchStudio-Reel exceeds author ground-truth posters on aesthetics (3.52 vs 2.94) and wins 84-93% of papers—is derived from a single run per paper evaluated by two VLM judges. The paper reports no standard deviation, confidence intervals, or seed variation for the generation process. In generative AI tasks, performance can fluctuate significantly based on random seeds, especially when using iterative loops. A 0.58 point margin on a 5-point scale with n=100 is plausible but not definitive without variance estimates. The reader cannot distinguish a robust effect from a lucky seed or a specific alignment between the VLM judges' preferences and the system's output style. Reporting results across multiple seeds (e.g., 3-5) with mean ± SD is essential to establish the stability of this gain.

Second, the ablation study in Section 5.1 attempts to isolate the contribution of the "skill machinery" (the measured-fill loop) by comparing a single-shot prompt against the full pipeline. However, the comparison is confounded: the single-shot baseline uses a fixed, static prompt (Appendix C), while the pipeline employs a multi-turn agent with tool use (e.g., reading the DOM, executing code). The observed gain (~0.76 points) could be driven by the agent's ability to use tools and iterate, rather than the specific "measured-fill" logic itself. To rigorously attribute the gain to the loop, the authors should control for the agent harness, perhaps by comparing a single-shot agent with tool access but no iterative fill loop, or by running the loop with a simpler agent.

Finally, the claim of surpassing human authors relies entirely on VLM judges. While the paper notes that VLMs are used to avoid human bias, VLMs themselves have known biases towards certain visual styles (e.g., clean layouts, specific color palettes) that may not align with human expert preferences for scientific posters. The "author ground-truth" posters are often bespoke and may contain idiosyncratic design choices that VLMs penalize. Without a human evaluation study (even a small one) to validate that the VLM scores correlate with human expert judgment, the claim of "surpassing" human authors remains an artifact of the evaluation metric rather than a proven improvement in quality.

Action items.

- **[science]** Table 1 claims ResearchStudio-Reel beats author ground-truth on aesthetics (3.52 vs 2.94) with no reported variance or seed count. A single run per paper cannot distinguish a real effect from seed noise or VLM idiosyncrasy. Report results across 3-5 seeds with mean ± SD or bootstrap CIs to confirm stability. (science)
- **[science]** Section 5.1 attributes the ~0.76 aesthetic gain to the 'measured-fill loop' by comparing single-shot vs. pipeline. However, the pipeline also uses multi-turn tool use while the baseline is static. This confounds the loop with the agent harness. Add a control with tool use but no iterative loop to isolate the loop's specific contribution. (science)
- **[science]** The claim of 'surpassing authors' relies solely on VLM judges, which may bias towards the system's consistent layout over human bespoke designs. Include a small human evaluation (n=20-30) blind to source to verify VLM scores correlate with expert human

preference before claiming superiority. (science)

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is currently insufficient to support the strength of the claims made regarding performance superiority. While the experimental design (100 papers, multiple baselines) is sound, the analysis of the resulting numbers lacks necessary rigor in three specific areas.

First, **uncertainty is missing**. Table 1 and the Appendix present point estimates (means) for aesthetic and information scores (e.g., 3.33, 4.00) derived from 100 papers. However, no standard deviation (SD), standard error (SE), or confidence intervals (CI) are reported. In generative AI tasks, variance across samples and judges is often high; without a measure of spread, a difference of 0.1 or 0.2 points is indistinguishable from noise. The reader cannot determine if the reported "win" is stable or a fluke of the specific 100-paper sample.

Second, **statistical significance is asserted without evidence**. The text repeatedly uses terms like "significantly better," "surpasses," and "leads" (Abstract; Section 5 Results). These terms imply a formal hypothesis test was conducted (e.g., a paired t-test or Wilcoxon signed-rank test comparing the proposed method against the author ground-truth or baselines on the same 100 papers). No such tests, p-values, or effect sizes are reported. The authors appear to be inferring significance solely from the magnitude of the mean difference, which is statistically invalid without a test of variance.

Third, **multiple comparisons are uncorrected**. The study evaluates 7 systems across 8 distinct metrics (Aesthetic sub-criteria, Information sub-criteria, Quiz splits), resulting in dozens of pairwise comparisons. The table highlights "best" and "second best" values without adjusting for the family-wise error rate. With ~50+ tests, one expects several false positives by chance alone. The lack of a correction method (Bonferroni, Holm, or Benjamini-Hochberg) inflates the risk of Type I errors in the reported "wins."

These are reporting gaps that can be fixed by re-analyzing the existing data (which the authors presumably have) rather than re-running the entire generation pipeline. The authors should compute and report SD/CI for all means, run paired statistical tests for the key comparisons, and apply a multiple-comparison correction to the benchmark results.

Action items.

- **[writing]** Table 1 reports mean scores (e.g., 3.33, 4.00) for 100 papers but omits uncertainty measures (SD, SE, or 95% CI). Report mean ± SD or 95% CIs for all quality metrics in Table 1 and the Appendix to allow assessment of stability.
- **[writing]** The abstract and Section 5 claim the method is 'significantly better' without

reporting a formal hypothesis test (e.g., paired t-test) or p-values. Either run paired tests on the 100 paper scores and report p-values, or rephrase claims to 'higher mean score' without invoking statistical significance.

- [**writing**] Table 1 compares 7 systems across 8 metrics (56 comparisons) and highlights 'best' values without multiple-comparison correction (e.g., Bonferroni or FDR). Apply a correction method to pairwise comparisons or explicitly state that 'best' labels are uncorrected and may include chance findings.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the narrative flow is strong, but several sentences are overly long and complex, forcing the reader to re-parse them to recover the intended meaning. The abstract contains a grammatical error and a run-on sentence that obscures the contrast between prior work and the proposed method. In Section 3.1, the description of the figure-cleanup chain is a single, dense run-on sentence that lists multiple steps without clear breaks; splitting this would improve readability. Similarly, the results analysis in Section 4 uses a long sentence to compare multiple systems and metrics, which dilutes the impact of the key finding. Finally, the use of 'load-bearing' as a metaphor in Section 5.1, while technically precise for the authors, may be jarring for a general academic audience and should be replaced with a more standard term like 'critical' or 'essential'. These are minor friction points in an otherwise clear manuscript.

Action items.

- [**writing**] The paper is generally well-structured and the narrative flow is strong, but several sentences are overly long and complex, forcing the reader to re-parse them to recover the intended meaning. The abstract contains a grammatical error and a run-on sentence that obscures the contrast between prior work and the proposed method. In Section 3.1, the description of the figure-cleanup chain is a single, dense run-on sentence that lists multiple steps without clear breaks; splitting this would improve